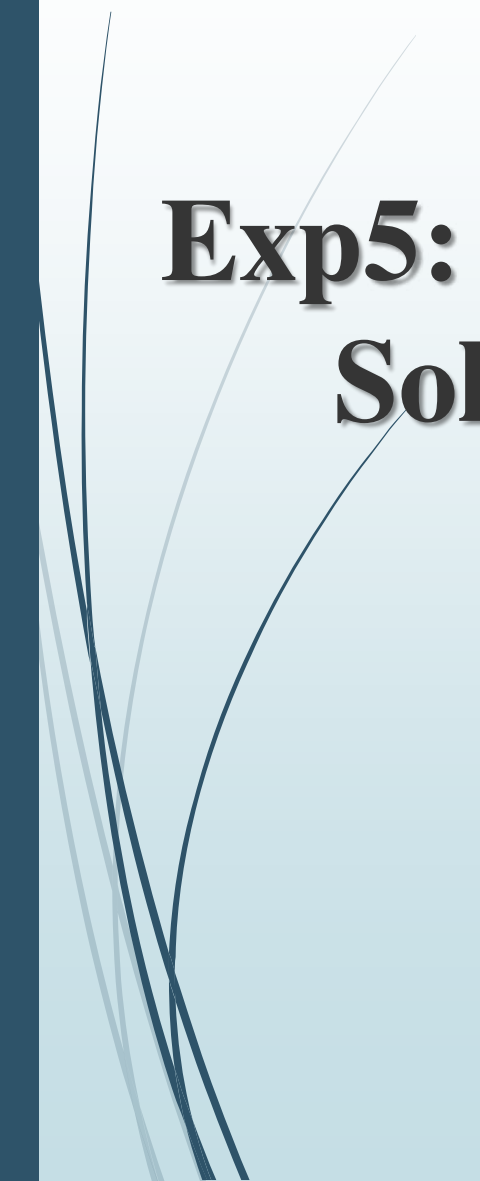




# **Exp5: Determination of Heat of Solution From Solubility**

# Theory

When a sufficiently large amount of solute is in contact with the limited amount of solvent, dissolving of the solute starts taking place. After a certain period of time, the reverse process, precipitation, starts to occur. When dissolving and precipitation of the solute take place at equal rate (equilibrium has been established), the solution is said to be saturated. After this point the concentration of the solute in the solution remaining unchanged. The concentration of the saturated solution is defined as the solubility of the solute in the given solvent. Solubility is generally a function of temperature.

# Theory

**Heat of Solution:-** is the amount of heat that absorbed when one mole of the solid is dissolved in a solution that is already practically saturated.

**Solubility:-** involves to measure the concentration of the solute in the saturated solution.

At equilibrium:

The chemical reactions are two kinds

1-Exothermic

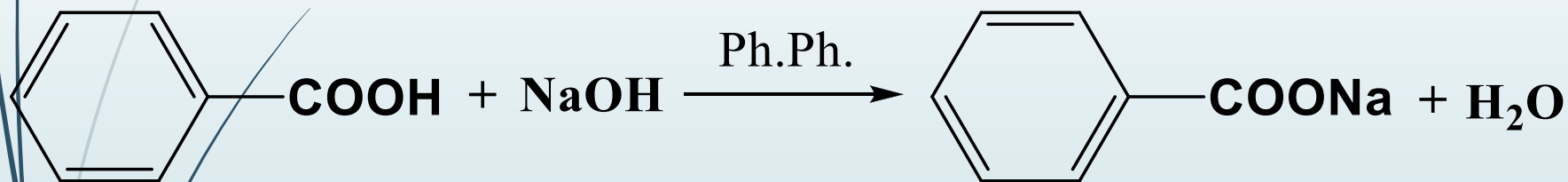
2-Endothermic

$$K_s = \frac{\text{dissolved solute conc.}}{\text{solid solute conc.}} \Rightarrow \frac{C_s}{C_s^*}$$

$$\begin{aligned} \Delta H (-) \quad C_s &\propto \frac{1}{T} \quad \text{Exothermic} \\ \Delta H (+) \quad C_s &\propto T \quad \text{Endothermic} \end{aligned}$$

\* The solubility depends on the temperature, pressure and nature of the solute & solvent.

\*\* The solubility of solute (benzoic acid) can be determined as a function of temperature. A plot of  $\ln C_s$  versus  $1/T$  should give straight line with the slop being equal to heat of solution.



# Vant Hoff equation

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -RT \ln K_s$$

$$-RT \ln K_s = \Delta H - T\Delta S$$

$$\ln K_s = \frac{\Delta H}{-RT} - \frac{T\Delta S}{-RT}$$

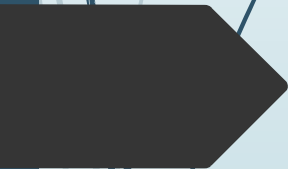
$$\ln K_s \text{ or } \ln C_s = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

Where  **$K_s$  or  $C_s$**  = Solubility  $\left(\frac{\text{mole}}{\text{Kg}} \text{ of solvent}\right)$

**$T$**  = Temperature ( $^{\circ}\text{F}$ )

**$\Delta H$**  = Heat of solution  $\left(\frac{\text{J}}{\text{mole}}\right)$

# Procedure

- 1-** Weight 1 gm of benzoic acid and add it to approximately 250ml of distilled water that has been heated up to 60°C in a beaker. Caution, above 70°C benzoic acid may be decompose.
  - 2-** Use the pipette to transfer 10 ml from the filtrate to the conical flask, and titrate with the standard (0.1N) NaOH solution, using phenol phthalein as the indicator.
  - 3-** Repeat step 2 at different temperature (30)°C.
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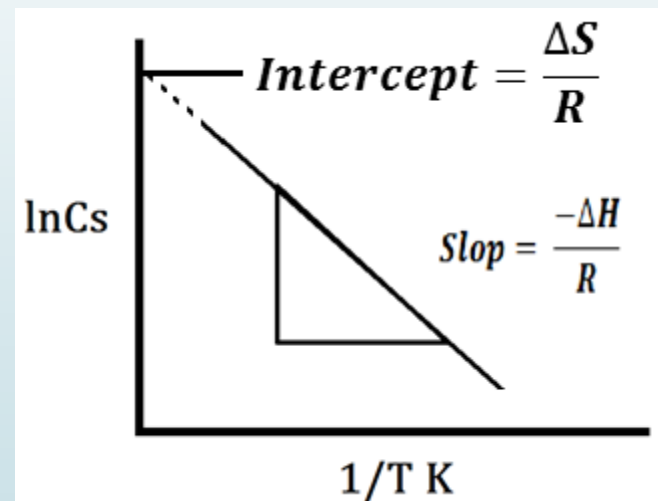
# Calculation

1- Complete the table below:

| T°C | TK | 1/TK | Vml<br>NaOH | Cs                                                                                       | lnCs |
|-----|----|------|-------------|------------------------------------------------------------------------------------------|------|
|     |    |      |             | Benzoic acid   NaOH<br>$N_1 * V_1 = N_2 * V_2$<br>$N_1 * 10 = 0.1 * \text{From burette}$ |      |

2- Draw the plot between lnCs versus 1/T.

3- Calculate the heat of solution from the plot.



# Discussion

**Q1/** What is the factors that affected on the solubility?

**Q2/** What is the Goal of the experiment?

**Q3/** What the differences between exothermic and endothermic reaction?